Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-E Primary Myocardial Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-05-C01				
NPLEX-P1-CARD-18-05-C02				
NPLEX-P1-CARD-18-05-C03				
NPLEX-P1-CARD-18-05-C04				
NPLEX-P1-CARD-18-05-C05				
NPLEX-P1-CARD-18-05-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C01-BD1

Rebuild dilated cardiomyopathy from its central defect

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-BD1 Date: _____

Without looking at your notes, write the central defect in dilated cardiomyopathy in one sentence, then draw the ventricle in cross section beside a normal ventricle and label cavity size and wall thickness in each. Write which direction sarcomeres are added, what that pattern of hypertrophy is called, and why total heart weight rises while the wall does not thicken. Write the law of Laplace as a relationship between wall stress, pressure, radius, and wall thickness, then use it to explain in three or four sentences why dilation keeps worsening once it starts. Finish by listing at least seven causes, and beside each one write how it injures myocytes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C01-BD2

Explain every finding in a large, weak ventricle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-BD2 Date: _____

A 46-year-old man has an echocardiogram showing all four chambers enlarged, a left ventricular wall of 8 mm, and an ejection fraction of 22%. He has a holosystolic murmur at the apex, a second holosystolic murmur at the left lower sternal border, an extra sound in early diastole, raised jugular venous pressure, and ankle edema. Explain each of those findings from the size and weakness of the ventricle, naming for the two murmurs why the valves leak even though the leaflets are structurally normal. Write what the extra diastolic sound is and the mechanical event that produces it. Then write the two complications that follow from blood sitting in a poorly contracting chamber and from stretched, fibrotic myocardium, and say why each one happens. Finish by writing what you would look for on imaging and in the history to separate this from ischemic cardiomyopathy, and name the feature of wall motion that differs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-05-C01-RT01

Rebuild the remodeling spiral

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-RT01 Date: _____

Draw the remodeling spiral of dilated cardiomyopathy as a closed loop with five steps, writing each step in order around the loop. Beside the step where the cavity enlarges, write the two separate processes that enlarge it and which neurohormonal change drives one of them. Beside the step on wall stress, write the Laplace relationship and mark which two variables are moving and in which direction. Beside the final step, write why higher wall stress costs the myocardium oxygen. Then write one sentence saying what happens to the ejection fraction with each turn of the loop.

Rebuild from memory: The remodeling spiral of dilated cardiomyopathy

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C02-BD1

Rebuild hypertrophic cardiomyopathy from gene to tissue to filling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C02 Activity: NPLEX-P1-CARD-18-05-C02-BD1 Date: _____

Write the inheritance pattern of hypertrophic cardiomyopathy, its approximate prevalence, and the two genes most often involved with the protein each one encodes. Write what the mutant sarcomeres do differently and why the muscle thickens with no pressure load on it. Draw a short axis view of the left ventricle showing where the hypertrophy is thickest and what has happened to the cavity, then list the three histological findings. Next write which half of the cardiac cycle is primarily impaired and why a thick, fibrotic wall produces that impairment, then write what happens to the ejection fraction and why. Finish by writing which extra heart sound appears, which part of filling the patient becomes dependent on, why atrial fibrillation is poorly tolerated, and why exertional angina occurs with normal epicardial coronary arteries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C02-BD2

Reason through obstruction, syncope, and sudden death

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C02 Activity: NPLEX-P1-CARD-18-05-C02-BD2 Date: _____

Rebuild the five-step sequence that turns septal thickening into dynamic outflow obstruction, then write in two or three sentences the single rule that decides whether the obstruction gets better or worse, stated in terms of ventricular cavity size and contractility. Apply that rule to a 17-year-old who has a louder murmur when he stands up quickly and a softer one when he squats, and explain each change. Describe the murmur itself, where it is loudest, whether it radiates to the carotids, and what the carotid pulse contour feels like. Then write why this disease is a leading cause of sudden death in young athletes, naming the tissue features that create the arrhythmia substrate and the three things exertion adds. Write the two separate mechanisms that can produce exertional syncope here. Finish with the four features that separate a young athlete with physiologic hypertrophy from this disease.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-05-C02-CP01

Hypertrophic cardiomyopathy against the athlete's heart

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Hypertrophic cardiomyopathy	Athlete's heart
What drives the hypertrophy		
Symmetry and site of wall thickening		
Left ventricular cavity size		
Diastolic filling		
What happens after a period of rest from training		
Tissue architecture of the myocardium		
Risk of exertional arrhythmia and sudden death		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-05-C02-PR01

Predict what the strain phase of Valsalva does in hypertrophic cardiomyopathy

A 24-year-old woman with hypertrophic cardiomyopathy and a resting outflow murmur is asked to bear down and hold it. Predict each item below for the strain phase, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return to the right heart		
What happens to left ventricular end-diastolic volume		
What happens to the distance between the anterior mitral leaflet and the septum		
What happens to the timing and extent of systolic anterior motion		
What happens to the loudness of the outflow murmur		
What happens to the loudness of a murmur of aortic stenosis under the same maneuver		
What happens to the murmur instead when she squats or her legs are raised passively		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C03-BD1

Rebuild restrictive cardiomyopathy and its causes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C03 Activity: NPLEX-P1-CARD-18-05-C03-BD1 Date: _____

Write what makes the ventricular wall stiff in restrictive cardiomyopathy, naming the two broad ways it happens. Define compliance in your own words, then draw a pressure against volume curve for filling in a normal ventricle and in a restrictive one on the same axes and label which is steeper. Draw a ventricular pressure tracing through diastole showing rapid early filling that stops abruptly, and write the name of that pattern. Write what happens to cavity size and ejection fraction and why. Then build a table of six causes with one row each, using the columns cause, what stiffens the wall, and clues, and fill every cell. Finish by writing which cause is most common and, for the eosinophilic cause, the three stages that run from the initial injury to the obliterated apex.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C03-BD2

Trace high filling pressures to the bedside and read the ECG mismatch

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C03 Activity: NPLEX-P1-CARD-18-05-C03-BD2 Date: _____

A 71-year-old man has a normal left ventricular cavity, a wall of 16 mm, an ejection fraction of 55%, and both atria markedly enlarged. Write why the atria are this large and the two problems that follow from atrial dilation. Write which side of the circulation tends to dominate his examination and list the findings you expect there, including what his jugular venous pressure does on inspiration and the name of that sign. Explain in two or three sentences why his cardiac output cannot rise much with exercise, referring to how stroke volume would normally be increased. His electrocardiogram shows low voltage in the limb leads. Rebuild the five-step sequence that explains low voltage despite a thick wall, and write what the same wall thickness would do to voltage if it were true muscular hypertrophy. Finish by writing which stain is used on the biopsy, what it shows under polarized light, and which other condition produces nearly this whole picture from outside the myocardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C04-BD1

Sort the three cardiomyopathies by which half of the cycle fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C04 Activity: NPLEX-P1-CARD-18-05-C04-BD1 Date: _____

Write the single question you ask first to separate the three cardiomyopathies, then answer it for dilated, hypertrophic, and restrictive disease in one sentence each, naming whether the dysfunction is systolic or diastolic and whether the ejection fraction is reduced or preserved. For the two diastolic diseases, write what makes the wall stiff in each. Then draw pressure against volume loops on one set of axes for a normal ventricle, a dilated one, a hypertrophic one, and a restrictive one. Label the end-systolic pressure-volume relationship and the end-diastolic pressure-volume relationship on each, mark which line has moved in which direction, and write beside each loop which property of the ventricle that movement represents. Finish by writing the one consequence all three diseases share and the tissue change that explains it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C04-BD2

Classify four sets of measurements

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C04 Activity: NPLEX-P1-CARD-18-05-C04-BD2 Date: _____

Four patients have echocardiography and an electrocardiogram. Patient A: left ventricular end-diastolic diameter enlarged, wall 7 mm, ejection fraction 20%, extra sound in early diastole. Patient B: septum 21 mm, posterior wall 10 mm, small cavity, ejection fraction 72%, anterior mitral leaflet touching the septum in systole. Patient C: normal cavity, wall 16 mm, ejection fraction 55%, both atria very large, low voltage in the limb leads. Patient D: wall 19 mm, tall QRS complexes, small cavity, ejection fraction 70%, deep narrow Q waves in the septal leads. For each patient write in order whether emptying or filling has failed, which cardiomyopathy it is, and the one measurement that settled it. For patients C and D explain why the same wall thickness gives opposite voltages. Then write what you expect each patient's pressure-volume loop to look like, and which extra heart sound each is most likely to have.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-05-C04-CP02

The three cardiomyopathies side by side

Fill in the table from memory on blank paper, then check it against the notes and correct any cell you missed.

Feature	Dilated	Hypertrophic	Restrictive
Ventricular cavity size			
Wall thickness			
Pattern of hypertrophy, if any			
Ejection fraction			
Primary dysfunction			
Position and shape of the pressure-volume loop			
Typical extra heart sound			
Characteristic electrocardiogram finding			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C05-BD1

Rebuild the causes and histology of myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-BD1 Date: _____

Define myocarditis in one sentence, including the word that excludes the commonest other cause of myocyte death. Write which group of organisms causes most cases in developed countries and name at least five of them. For the classic virus, write the receptor it uses to enter myocytes, which other virus shares that receptor, and which structural protein a viral protease cleaves. Write the two separate processes that injure myocytes in viral disease. Then build a table of seven causes with one row each, using the columns cause and mechanism with distinguishing features, and fill every cell, including the vector or route where the notes gave one. Finish by writing the defining histological finding and the name of those criteria, then describe the infiltrate in viral disease, in hypersensitivity disease, and in giant cell disease, and write why a negative biopsy does not exclude the diagnosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C05-BD2

Explain the findings of myocarditis and its late outcome

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-BD2 Date: _____

A 23-year-old man has chest pain and palpitations ten days after a febrile illness. Troponin is raised, the coronary arteries are normal, the electrocardiogram shows diffuse ST elevation, and the ejection fraction is 35%. Explain each of those four findings from the lesion in the myocardium, saying for troponin why it rises without coronary occlusion, for the rhythm disturbance which two electrical mechanisms the inflammation creates, and for the low ejection fraction what has been lost. Write why his chest pain could be either ischemic in character or pleuritic, and what the second pattern implies about which layer is involved. Then rebuild the six-step sequence from viral entry to dilated cardiomyopathy and write which step makes the damage permanent. Finish with three patients: one has many eosinophils with rash and fever, one has multinucleated giant cells in widespread necrosis, and one has parasites inside myocytes. For each, write the cause and the one feature of the history or tissue that named it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-05-C05-RT02

Rebuild the path from viral infection to dilated cardiomyopathy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-RT02 Date: _____

Rebuild the six-step sequence from viral entry into the myocyte to established dilated cardiomyopathy, one step per line and in order. Beside step one write the receptor and the two viruses that use it. Beside the immune steps write which cell does the killing and how exposed cardiac proteins can keep the process going after the virus is cleared. Beside the final two steps write what replaces the lost myocytes and the one mechanical property that tissue does not have. Then write one sentence saying why the ejection fraction falls as a result.

Rebuild from memory: From viral infection to dilated cardiomyopathy

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-05-C05-CA01

A young man with chest pain, a raised troponin, and clean coronary arteries

A 26-year-old man comes to the emergency department because of 2 days of chest pain, breathlessness on climbing stairs, and the sensation of his heart skipping. Twelve days earlier he had fever, sore throat, and loose stools that settled without treatment. Temperature is 37.8 degrees Celsius (100.0 degrees Fahrenheit), pulse is 108/min, respirations are 22/min, and blood pressure is 102/64 mm Hg. Jugular venous pressure is raised and there is a soft extra sound in early diastole. The electrocardiogram shows ST elevation in the anterior, lateral, and inferior leads with occasional wide early beats, and intermittent delay between the atria and the ventricles. Serum troponin is markedly raised and C-reactive protein is high. Echocardiography shows a left ventricle of normal cavity size with diffusely reduced wall motion and an ejection fraction of 34%. Coronary angiography shows no obstructive disease. An endomyocardial biopsy shows a patchy interstitial infiltrate of lymphocytes surrounding injured myocytes, with no eosinophils, no giant cells, and no granulomas.

1. Recognize. Name the condition, the most likely class of cause given the preceding illness, and the single organism most classically responsible.

2. Explain. Explain the raised troponin, the diffusely reduced wall motion, the atrioventricular delay, and the wide early beats from what the inflammation is doing to myocytes and to conduction.

3. Differentiate. Write how this differs from acute myocardial infarction, from stress cardiomyopathy, and from giant cell and hypersensitivity myocarditis, naming for each the finding on this patient's results that separates it.

4. Investigate. Write which two features on the biopsy you would name to satisfy the defining histological criteria, and explain why a biopsy without them would not exclude the diagnosis.

5. Decide. State what this illness can progress to over months to years if myocytes are lost, and what the ventricle will look like if it does.

6. Justify. Justify that answer by writing the sequence from viral entry to that end state, naming the receptor involved, the immune mechanism, what replaces lost myocytes, and why the law of Laplace keeps the process going once the ventricle dilates.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C06-BD1

Rebuild stress and tachycardia-induced cardiomyopathy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C06 Activity: NPLEX-P1-CARD-18-05-C06-BD1 Date: _____

Write the two other names for stress cardiomyopathy, the patient group it mostly affects, and the kinds of trigger that set it off. Draw the left ventricle in systole in this condition, marking which segments are akinetic and which contract vigorously, and write the name of the pattern and what the shape is named after. Then rebuild the six-step proposed mechanism in order, including which catecholamine matters most, where its receptors are dense, which signaling change occurs at those receptors, and what happens to calcium in the myocyte. Separately, write what tachycardia-induced cardiomyopathy is and list at least three rhythms that cause it. Write what a sustained fast rate does to diastole and name the two things diastole is needed for, then write the four myocardial changes that follow and what the ventricle ends up looking like. Finish by writing the one feature both of these conditions share that sets them apart from the three classic cardiomyopathies.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C06-BD2

Separate a stunned ventricle from an infarcted one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C06 Activity: NPLEX-P1-CARD-18-05-C06-BD2 Date: _____

A 70-year-old woman has chest pain and breathlessness an hour after being told of her sister's death. The electrocardiogram shows anterior ST elevation with deep T wave inversion and a long QT interval, troponin is mildly raised, the apex and mid ventricle are akinetic with a vigorously contracting base, and the coronary arteries are normal. Write the four features of this presentation that argue against infarction as the cause, including how the size of the troponin rise compares with the amount of muscle not moving and why the akinetic area does not fit a coronary territory. Write what a biopsy would show and what that pattern tells you about the injury. Predict what her ventricular function will be in six weeks and explain the cellular reason, then write what the same sized infarct would have left behind. List the four complications that follow from an acutely weak apex and a hypercontracting base, saying for each how the geometry or the weakness produces it. Finish with a second patient whose ejection fraction is 30% during six months of a fast irregular rhythm and 55% two months after the rate is controlled. Write what that recovery tells you about the cause and why rate alone can lower the ejection fraction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-05-C02-Q01

A systolic murmur that changes with position

A 19-year-old man is evaluated after collapsing during a basketball game. His father died suddenly at age 34. Temperature is 37.0 degrees Celsius (98.6 degrees Fahrenheit), pulse is 78/min, and blood pressure is 118/72 mm Hg. A harsh crescendo-decrescendo systolic murmur is heard at the left lower sternal border that does not radiate to the carotids. The murmur becomes louder when he stands up from a squatting position and softer during sustained handgrip. An extra sound is heard just before the first heart sound. Echocardiography shows a septum of 22 mm, a posterior wall of 11 mm, a small left ventricular cavity, and an ejection fraction of 74%. Which of the following best explains why the murmur becomes louder on standing?

A. Reduced afterload increases the pressure gradient across a fixed valvular narrowing

B. Increased venous return raises end-diastolic volume and accelerates flow across the outflow tract

C. Reduced left ventricular volume brings the anterior mitral leaflet closer to the hypertrophied septum

D. Reduced contractility lengthens ejection time and prolongs turbulent flow

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-05-C03-Q02

A thick wall with small complexes

A 74-year-old man comes to the physician because of 4 months of breathlessness on exertion, abdominal swelling, and leg edema. Jugular venous pressure is raised and rises further on inspiration. The liver is enlarged and there is ascites. Echocardiography shows a left ventricular wall thickness of 17 mm with a speckled appearance, a normal left ventricular cavity, an ejection fraction of 54%, and marked enlargement of both atria. The electrocardiogram shows low QRS voltage in the limb leads. Which of the following best explains the combination of a thick wall with low QRS voltage?

A. Extracellular protein deposits add wall thickness without adding electrically active myocytes

B. Myocyte disarray disperses depolarization wavefronts so that their voltages cancel

C. Series addition of sarcomeres lengthens myocytes without increasing their number

D. Pericardial thickening attenuates the voltages recorded at the body surface

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-05-C06-Q03

Akinesis beyond one coronary territory

A 68-year-old woman comes to the emergency department because of chest pressure and breathlessness that began 2 hours after her husband's funeral. Pulse is 104/min and blood pressure is 96/60 mm Hg. The electrocardiogram shows ST elevation in leads V2 to V6 with a QT interval corrected for rate of 0.49 s. Serum troponin is mildly raised. Echocardiography shows akinesis of the apex and mid ventricle with vigorous contraction of the basal segments and an ejection fraction of 35%. Coronary angiography shows no obstructive disease. Which of the following findings would most likely be present on endomyocardial biopsy?

A. Interstitial infiltrate rich in eosinophils with myocyte necrosis

B. Extracellular material staining with Congo red and showing apple-green birefringence

C. Coagulative necrosis with dense neutrophil infiltration in a wedge-shaped region

D. Contraction band necrosis of myocytes without occlusive vascular disease

Why did you pick it?